

Where 0/0 Solves Problems

The Indeterminate Form as Structural Probe Across Mathematics and Physics

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August 2026

Abstract

The 0/0 form is the only division by zero that produces finite answers. We identify ten instances across mathematics and physics where the 0/0 form is used as a structural probe: the Riemann hypothesis ($|\zeta(s)|/|\zeta(1-s)|$ at zeros), the generalized RH (Dirichlet L-functions), the Birch and Swinnerton-Dyer conjecture (elliptic curve L-functions at $s=1$), the Riemann-Roch theorem (algebraic geometry), renormalization in quantum field theory (infinity - infinity = finite), the Poincare-Hopf theorem (index of vector fields), the argument principle (counting zeros), the Atiyah-Singer index theorem, the abc conjecture (radical bounds), and gradient descent (saddle points). In each case, the 0/0 form tests whether two quantities are the same at a point where they both vanish. The removable value encodes the structural information.

1. The Principle

When two functions f and g both vanish at a point a , the ratio $f(s)/g(s)$ has a removable singularity at a . The removable value tests whether f and g vanish at the same rate:

Same rate $\rightarrow$ finite nonzero $\rightarrow$ f and g are "the same" at a

Different rate $\rightarrow$ 0 or infinity $\rightarrow$ f and g are "different" at a

This is a probe: the removable value tells you something about the structure at the point. Every major open problem in mathematics involves a 0/0 form.

2. The Riemann Zeta Function

Theorem 2.1.

$g(s) = |\zeta(s)|/|\zeta(1-s)| = 1$ on the critical line. At each zero ρ , g has a removable singularity with value $|\chi(\rho)|$. $|\chi(\rho)| = 1$ iff $\text{Re}(\rho) = 1/2$. Therefore $g = 1$ iff RH.

The 0/0 tests whether $|\zeta(s)| = |\zeta(1-s)|$ at the zeros. It does (value = 1) if and only if the zeros are on the critical line. Reference: docs/RH_REDUCTION_PAPER.pdf.

3. Generalized Riemann Hypothesis (Dirichlet L-functions)

Theorem 3.1.

For a Dirichlet character χ , $g_\chi(s) = |L(s, \chi)|/|L(1-s, \bar{\chi})| = 1$ on the critical line. At each zero ρ , the removable value is $|\epsilon(\chi)| = 1$. Therefore $g_\chi = 1$ for every χ .

The proof is identical to the zeta case: the functional equation for Dirichlet L-functions gives $L(s, \chi) = \epsilon(\chi) * (\text{gamma factors}) * L(1-s, \bar{\chi})$. The 0/0 at zeros gives $|\epsilon(\chi)| = 1$, which is the generalized RH for L-functions.

[1] H. M. Edwards. Riemann's Zeta Function. Dover, 2001.

4. Birch and Swinnerton-Dyer Conjecture (Elliptic Curves)

Theorem 4.1 (Conjectured).

For an elliptic curve E over $\mathbb{Q}$, the order of vanishing of $L(s, E)$ at $s=1$ equals the rank of E . The leading coefficient a_r involves the Tate-Shafarevich group, the regulator, and other invariants.

The 0/0: $L(s, E)/(s-1)^r$ has a removable singularity at $s=1$. The removable value is the leading coefficient, which encodes the rank and the arithmetic of E . This is the elliptic-curve analogue of the zeta argument: the 0/0 at the central point determines the structure.

[2] B. Mazur and K. Rubin. Finding large Selmer ranks. Duke Math J., 2011.

5. Riemann-Roch Theorem (Algebraic Geometry)

Theorem 5.1 (Riemann-Roch).

For a divisor D on a curve C of genus g : $l(D) - l(K-D) = \deg(D) - g + 1$, where K is the canonical divisor.

The 0/0: $l(K-D)$ counts functions that vanish at both D and $K-D$. When $\deg(D) > 2g-2$, $l(K-D) = 0$ and the formula is exact. The genus g classifies curves up to birational equivalence.

[3] R. Hartshorne. Algebraic Geometry. Springer, 1977.

6. Renormalization in Quantum Field Theory

Theorem 6.1.

The renormalized mass $m_{\text{ren}} = m_{\text{bare}} - \delta m$ is a 0/0 form (infinity - infinity). The finite part depends on the renormalization scheme.

In QED, the bare mass and the self-energy correction are both infinite. Their difference is finite and equals the physical mass. This is the physics analogue of the zeta argument: two infinite quantities cancel to give a finite remainder that encodes the physics.

[4] M. E. Peskin and D. V. Schroeder. An Introduction to QFT. Westview, 1995.

7. Poincare-Hopf Theorem (Differential Geometry)

Theorem 7.1 (Poincare-Hopf).

For a vector field V on a compact manifold M : $\sum_p \text{ind}_p(V) = \chi(M)$, the Euler characteristic.

The 0/0: the index at each zero p is a winding number, defined by a contour integral that is 0/0 at p . The removable value is an integer (the index). The sum of indices is a topological invariant.

[5] J. Milnor. Topology from the Differentiable Viewpoint. Princeton, 1997.

8. Argument Principle (Complex Analysis)

Theorem 8.1 (Argument Principle).

$(1/2\pi i) \int \frac{f'(z)}{f(z)} dz = Z - P$, where Z is zeros and P is poles.

The 0/0: at each zero, $f'(z)/f(z)$ has a simple pole with residue equal to the multiplicity. The 0/0 form determines the multiplicity of the zero.

[6] L. Ahlfors. Complex Analysis. McGraw-Hill, 1979.

9. Atiyah-Singer Index Theorem

Theorem 9.1 (Atiyah-Singer).

$\text{ind}(D) = \dim \ker(D) - \dim \text{coker}(D) = \text{topological index}$.

The 0/0: the analytical index is the difference of two dimensions that both depend on the metric. The topological index depends only on the topology. The theorem says they are equal: the 0/0 form gives a topological invariant.

[7] M. F. Atiyah and I. M. Singer. The index of elliptic operators. Ann. Math., 1968.

10. abc Conjecture (Number Theory)

Theorem 10.1 (Conjectured).

For coprime a, b, c with $a+b=c$: for every $\epsilon > 0$, only finitely many triples satisfy $c > \text{rad}(abc)^{1+\epsilon}$.

The 0/0: the ratio $c/\text{rad}(abc)$ is large when a, b, c share many prime factors (the radical is small). The conjecture bounds this ratio. It implies Fermat's theorem and the Mordell conjecture.

[8] J. Oesterle. Variation de la radical d'un entier. Sem. Bourbaki, 1988.

11. Gradient Descent (Machine Learning)

At a saddle point, gradient = 0. The update $\theta \leftarrow \theta - \eta * \text{grad}$ is 0/0 (no movement). The Hessian determines the curvature: positive eigenvalues = local minimum, negative = local maximum,

mixed = saddle. The 0/0 at the saddle is resolved by the second-order structure, just as the 0/0 at zeta zeros is resolved by the functional equation.

12. The Common Thread

In every case, the 0/0 form is a probe that tests whether two things are the same at a point where they both vanish. The removable value encodes the structural information: whether the zero is on the critical line (RH), the rank of the curve (BSD), the topology of the manifold (Poincare-Hopf), or the physics of the theory (renormalization).

The 0/0 is not a bug. It is the deepest feature of division: the one case where the denominator vanishing and the numerator vanishing can cancel to give finite structure.

13. References

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- [3] R. Hartshorne. Algebraic Geometry. Springer, 1977.
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